

Linear Regression

We know a thing when we understand it.

George Berkeley (British philosopher for whom the California city was named)

You can observe a lot just by watching.

Yogi Berra

We study the simple linear regression model which, except for random error, assumes a straight-line relationship between a response and an input variable. We use the method of least squares to estimate the parameters of this model. Assuming that the random error is normal with mean 0 and variance σ^2 , we show how to test hypotheses concerning the parameters of the model. The concept of regression to the mean is introduced; we explain when it arises and how one must be careful to avoid the regression fallacy in its presence. We explain the coefficient of determination. Finally, we introduce the multiple linear regression model, which relates a response variable to a set of input variables.

It was a spring day in 1888, and Francis Galton was out for a stroll in the countryside. While walking, he considered a question that had concerned him for some time. What was the relationship between a child's physical and mental characteristics and those of the child's parents? For instance, simplifying his ideas somewhat, Galton believed that the height of a child at adulthood should have an expected value equal to the height of his or her (same-sex) parent. But if this were so, then it would follow that about one-half of the offspring of very tall (short) people would be even taller (shorter) than their parents. Thus each new generation should produce taller (as well as shorter) people than the previous generation. However, on the contrary, data indicated a stability in the heights of the population from generation to generation. How could this apparent contradiction be explained?

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It came to Galton in a flash. In his own words, “A temporary shower drove me to seek refuge by a recess in the rock by the side of a pathway. There the idea flashed across me and I forgot everything else for a moment in my great delight.”

Galton’s flash of insight was that the mean value of a child’s characteristic (such as height) was not equal to his or her parent’s height but rather was between this value and the average value of the entire population. Thus, for instance, the heights of the offspring of very tall people (called, by Galton, people “taller than mediocrity”) would tend to be shorter than their parents. Similarly, the offspring of those shorter than mediocrity would tend to be taller than their parents. Galton called this insight “regression to mediocrity”; we call it *regression to the mean*.

12.1 INTRODUCTION

We are often interested in trying to determine the relationship between a pair of variables. For instance, how does the amount of money spent in advertising a new product relate to the first month’s sales figures for that product? Or how does the amount of catalyst employed in a scientific experiment relate to the yield of that experiment? Or how does the height of a father relate to that of his son?

In many situations the values of the variables are not determined simultaneously in time; rather, one of the variables will be set at some value, and this will, in turn, affect the value of the second variable. For instance, the advertising budget would be set before the sales figures are determined, and the amount of catalyst to be used would be set before the resulting yield could be determined. The variable whose value is determined first is called the *input* or *independent* variable and the other is called the *response* or *dependent* variable.

Suppose that the value of the independent variable is set to equal x . Let Y denote the resulting value of the dependent variable. The simplest type of relationship between this pair of variables is a straight-line, or *linear*, relation of the form

$$Y = \alpha + \beta x \quad (12.1)$$

This model, however, supposes that (once the values of the parameters α and β are determined) it would be possible to predict exactly the response for any value of the input variable. In practice, however, such precision is almost never attainable, and the most that one can expect is that the preceding equation is valid *subject to random error*.

In Sec. 12.2 we explain precisely the meaning of the *linear regression* model, which assumes that Eq. (12.1) is valid, subject to random error. In Sec. 12.3 we show how data can be used to estimate the regression parameters α and β .

The estimators presented are based on the least-squares approach to finding the best straight-line fit for a set of data pairs. Section 12.4 deals with the *error* random variable, which will be taken to be a normal random variable having mean 0 and variance σ^2 . The problem of estimating σ^2 will be considered.

In Sec. 12.5 we consider tests of the statistical hypothesis that there is no linear relationship between the response variable Y and the input value x . Section 12.6 deals with the concept of *regression to the mean*. It is shown that this phenomenon arises when the value of the regression parameter β is between 0 and 1. We explain how this phenomenon will often occur in testing–retesting situations and how a careless analysis of such data can often lead one into the *regression fallacy*. In addition, we indicate in this section how regression to the mean, in conjunction with the central limit theorem and the passing of many generations, can be used to explain why biological data sets are so often approximately normally distributed.

Section 12.7 is concerned with determining an interval that, with a fixed degree of confidence, will contain a future response corresponding to a specified input. These intervals, which make use of previously obtained data, are known as *prediction intervals*. Sections 12.8 and 12.9 present, respectively, the coefficient of determination and the correlation coefficient. Both quantities can be used to indicate the degree of fit of the linear regression model to the data. An approach to assessing the validity of the linear regression model, by analyzing the residuals, is dealt with in Sec. 12.10.

In Sec. 12.11 we consider the multiple linear regression model, where one tries to predict a response not on the basis of the value of a single input variable but on the basis of the values of two or more such variables.

12.2 SIMPLE LINEAR REGRESSION MODEL

Consider a pair of variables, one of which is called the *input variable* and the other the *response variable*. Suppose that for a specified value x of the input variable the value of the response variable Y can be expressed as

$$Y = \alpha + \beta x + e$$

The quantities α and β are parameters. The variable e , called the *random error*, is assumed to be a random variable having mean 0.

Definition. *The relationship between the response variable Y and the input variable x specified in the preceding equation is called a simple linear regression.*

The simple linear regression relationship can also be expressed by stating that for any value x of the input variable, the response variable Y is a random variable with mean given by

$$E[Y] = \alpha + \beta x$$

Thus a simple linear regression model supposes a straight-line relationship between the mean value of the response and the value of the input variable. Parameters α and β will almost always be unknown and will have to be estimated from data.

To see if a simple linear regression might be a reasonable model for the relationship between a pair of variables, one should first collect and then plot data on the paired values of the variables. For instance, suppose there is available a set of data pairs (x_i, y_i) , $i = 1, \dots, n$, meaning that when the input variable was set to equal x_i , the observed value of the response variable was y_i . These points should then be plotted to see if, subject to random error, a straight-line relationship between x and y appears to be a reasonable assumption. The resulting plot is called a *scatter diagram*.

■ Example 12.1

A new type of washing machine was recently introduced in 11 department stores. These stores are of roughly equal size and are located in similar types of communities. The manufacturer varied the price charged in each store, and the following data, giving the number of units sold in 1 month for each of the different prices, resulted.

Price (\$)	Number sold
280	44
290	41
300	34
310	38
320	33
330	30
340	32
350	26
360	28
370	23
380	20

A plot of the number of units sold y versus the price x for these 11 data pairs is given in Fig. 12.1. The resulting scatter diagram indicates that, subject to random error, the assumption of a straight-line relationship between the number of units sold and the price appears to be reasonable. That is, a simple linear regression model appears to be appropriate. ■

Remark. The input variable x in the linear regression model is not usually thought of as being a random variable. Rather it is regarded as a constant that can be set at various values. The resulting response Y , on the other hand, is regarded as a random variable whose mean value depends in a linear way on the input x . It is for this reason that we use the capital, or uppercase, letter Y to represent the response. We use a small, or lowercase, y to denote an observed value of Y , and so y will represent the observed value of the response at the input value x .

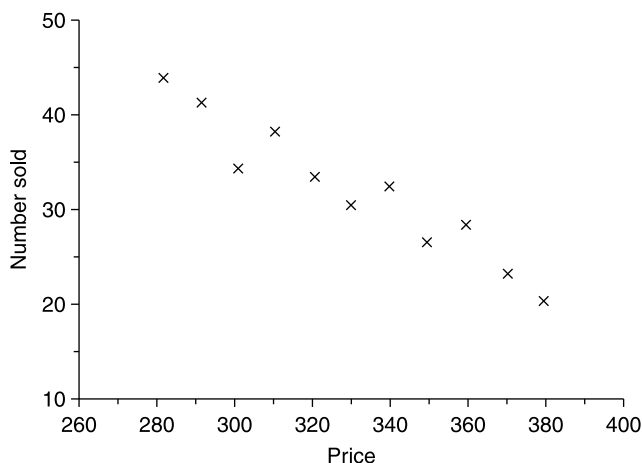


FIGURE 12.1 A scatter diagram for the data of Example 12.1.

PROBLEMS

- The following 12 data pairs relate y , the percentage yield of a laboratory experiment, to x , the temperature at which the experiment was conducted.

i	x_i	y_i	i	x_i	y_i
1	100	45	7	150	69
2	110	51	8	160	74
3	120	54	9	170	78
4	125	53	10	180	86
5	130	59	11	190	89
6	140	63	12	200	94

- Represent these data in a scatter diagram.
 - Do you think a simple linear regression model would be appropriate for describing the relationship between percentage yield and temperature?
- An area manager in a department store wants to study the relationship between the number of workers on duty and the value of merchandise lost to shoplifters. To do so, she assigned a different number of clerks for each of 10 weeks. The results were as follows:

Week	Number of workers	Loss
1	9	420
2	11	350
3	12	360
4	13	300
5	15	225
6	18	200
7	16	230
8	14	280
9	12	315
10	10	410

- (a) Which variable should be the input variable and which should be the response?
 - (b) Plot the data in a scatter diagram.
 - (c) Does a simple linear regression model appear reasonable?
3. The following data relate the traffic density, described in the number of automobiles per mile, to the average speed of traffic on a moderately large city thoroughfare. The data were collected at the same location at 10 different times within a span of 3 months.

Density	Speed
69	25.4
56	32.5
62	28.6
119	11.3
84	21.3
74	22.1
73	22.3
90	18.5
38	37.2
22	44.6

- (a) Which variable is the input and which is the response?
 - (b) Draw a scatter diagram.
 - (c) Does a simple linear regression model appear to be reasonable?
4. Repeat Prob. 3, but now let the square root of the speed, rather than the speed itself, be the response variable.
5. The use that can be obtained from a tire is affected by the air pressure in the tire. A new type of tire was tested for wear at different pressures, with the following results:

Pressure (pounds per square inch)	Mileage (thousands of miles)
30	29.4
31	32.2
32	35.9
33	38.4
34	36.6
35	34.8
36	35.0
37	32.2
38	30.5
39	28.6
40	27.4

- Plot the data in a scatter diagram.
- Does a simple linear regression model appear appropriate for describing the relation between tire pressure and miles of use?

12.3 ESTIMATING THE REGRESSION PARAMETERS

Suppose that the responses Y_i corresponding to the input values x_i , $i = 1, \dots, n$, are to be observed and used to estimate the parameters α and β in a simple linear regression model

$$Y = \alpha + \beta x + e$$

To determine estimators of α and β , we reason as follows: If A and B were the respective estimators of α and β , then the estimator of the response corresponding to the input value x_i would be $A + Bx_i$. Since the actual response is Y_i , it follows that the difference between the actual response and its estimated value is given by

$$\epsilon_i \equiv Y_i - (A + Bx_i)$$

That is, ϵ_i represents the error that would result from using estimators A and B to predict the response at input value x_i (Fig. 12.2).

Now, it is reasonable to choose our estimates of α and β to be the values of A and B that make these errors as small as possible. To accomplish this, we choose A and B to minimize the value of $\sum_{i=1}^n \epsilon_i^2$, the sum of the squares of the errors. The resulting estimators of α and β are called *least-square estimators*.

Definition. For given data pairs (x_i, Y_i) , $i = 1, \dots, n$, the least-square estimators of α and β are the values of A and B that make

$$\sum_{i=1}^n \epsilon_i^2 = \sum_{i=1}^n (Y_i - A - Bx_i)^2$$

as small as possible.

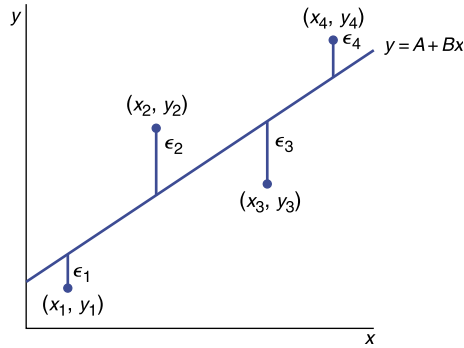


FIGURE 12.2 The errors.

Remark. The reason we want $\sum_{i=1}^n \epsilon_i^2$, rather than $\sum_{i=1}^n \epsilon_i$, to be small is that the sum of the errors can be small even when individual error terms are large (since large positive and large negative errors cancel). On the other hand, this could not happen with the sum of the **squares** of the errors since none of the terms could be negative.

It can be shown that the least-squares estimators of α and β , which we call $\hat{\alpha}$ and $\hat{\beta}$, are given by

$$\hat{\beta} = \frac{\sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\alpha} = \bar{Y} - \hat{\beta}\bar{x}$$

where

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} \quad \text{and} \quad \bar{Y} = \frac{\sum_{i=1}^n Y_i}{n}$$

The line

$$y = \hat{\alpha} + \hat{\beta}x$$

is called the *estimated regression line*: $\hat{\beta}$ is the slope, and $\hat{\alpha}$ is the intercept of this line.

Notation: If we let

$$S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y})$$

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$$

$$S_{YY} = \sum_{i=1}^n (Y_i - \bar{Y})^2$$

then the least-squares estimators can be expressed as

$$\hat{\beta} = \frac{S_{XY}}{S_{XX}}$$

$$\hat{\alpha} = \bar{Y} - \hat{\beta}\bar{x}$$

The values of $\hat{\alpha}$ and $\hat{\beta}$ can be obtained either by a pencil-and-paper computation or by using a hand calculator. In addition, Program 12-1 will compute the least-squares estimators and the estimated regression line. This program also gives the user the option of computing some other statistics whose values will be needed in the following sections.

■ Example 12.2

A large midwestern bank is planning on introducing a new word processing system to its secretarial staff. To learn about the amount of training that is needed to effectively implement the new system, the bank chose eight employees of roughly equal skill. These workers were trained for different amounts of time and were then individually put to work on a given project. The following data indicate the training times and the resulting times (both in hours) that it took each worker to complete the project.

Worker	Training time (= x)	Time to complete project (= Y)
1	22	18.4
2	18	19.2
3	30	14.5
4	16	19.0
5	25	16.6
6	20	17.7
7	10	24.4
8	14	21.0

- What is the estimated regression line?
- Predict the amount of time it would take a worker who receives 28 hours of training to complete the project.
- Predict the amount of time it would take a worker who receives 50 hours of training to complete the project.

Solution

- Rather than calculating by hand (which you will be asked to do in Prob. 2), we run Program 12-1, which computes the least-squares estimators and related statistics in simple linear regression models. We obtain the following:

First, enter the number of data pairs n , which is 8.

Next, enter the 8 successive pairs, which are:

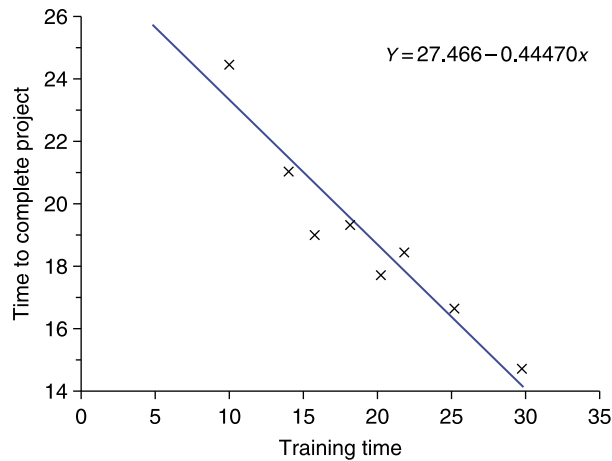


FIGURE 12.3 A scatter diagram and the estimated regression line.

22, 18.4
 18, 19.2
 30, 14.5
 16, 19
 25, 16.6
 20, 17.7
 10, 24.4
 14, 21

The program computes the least-squares estimators as follows:

$$A = 27.46606$$

$$B = -0.4447002$$

The estimated regression line is as follows:

$$Y = 27.46606 - 0.4447002x$$

A plot of the scatter diagram and the resulting estimated regression line is given in Fig. 12.3.

- (b) The best prediction of the completion time corresponding to the training time of 28 hours is its mean value, namely,

$$\alpha + 28\beta$$

By using the estimates of α and β previously derived, the predicted completion time is

$$27.466 - 28(0.445) = 15.006$$

- (c) This part asks for the prediction at the input value 50, which is far greater than all the input values in our data set. As a result, even though the

scatter diagram indicates that a straight-line fit should be a reasonable approximation for the range of input values considered, one should be extremely cautious about assuming that the relationship will continue to be a straight line for input values as large as 50. Thus, it is prudent not to attempt to answer part (c) on the basis of the available data. ■

Warning: Do not use the estimated regression line to predict responses at input values that are far outside the range of the ones used to obtain this line.

The following formulas can be useful when you are computing by hand.

$$S_{XY} = \sum_{i=1}^n x_i Y_i - n\bar{x}\bar{Y}$$

$$S_{XX} = \sum_{i=1}^n x_i^2 - n\bar{x}^2$$

PROBLEMS

- Find, by a hand computation, the estimated regression line for the following data:

x	y
1	4
2	7
3	8
5	12

- Plot the scatter diagram, and draw the estimated regression line.
 - Double all the data values and repeat part (a).
- Verify the value given in [Example 12.2](#) for the estimated regression line either by a pencil-and-paper computation or by using a hand calculator.
 - The following pairs of data represent the amounts of damages (in thousands of dollars) in fires at middle-class residences in a certain city and the distances (in miles) from these residences to the nearest fire station.

Distance	Damage
5.2	36.2
4.4	28.8
3.0	22.6
1.2	8.8
7.5	41.5
9.4	25.4

- Draw a scatter diagram.
- Try to approximate the relationship between the distance and damage by drawing a straight line through the data.

- (c) Find the estimated regression line, and compare it to the line drawn in part (b).
4. Consider Prob. 1 of Sec. 12.2.
- (a) Draw a straight line through the data points.
- (b) Determine the estimated regression line, and compare it to the line drawn in part (a).
5. The amounts (in millions of pounds) of poultry products consumed in the United States for the years 1995 through 2002 are as follows:
- 25.9 26.8 27.3 27.8 29.6 30.5 30.8 32.6
- (a) Letting the year be the independent variable and consumption the dependent variable, plot a scatter diagram.
- (b) Find the estimated regression line. (To simplify the algebra, you can take the x variable to be the year minus 1995. That is, year 1985 has value 0, 1996 has value 1, and so on.)
- (c) Plot the estimated regression line on the scatter diagram.
- (d) Predict the 1994 consumption figure.
- (e) Predict the 2004 consumption figure.
6. The following are the average 2003 math SAT scores in a sample of states, along with the percentage of graduating seniors who took the test.

State	Average score	Percentage
Arizona	525	38
California	519	54
Indiana	504	63
Missouri	583	8
Louisiana	559	8
Oregon	527	57
Virginia	510	71
Wisconsin	594	7
Texas	491	57
Vermont	512	70

Find the estimated regression line.

7. The following table relates the world production of wood pulp to the world production of newsprint for each of seven different years. The data come from the Statistical Division of the United Nations, New York, *Monthly Bulletin of Statistics* and are in units of 1 million metric tons.

Wood pulp	Newsprint
124.4	25.4
131.3	27.8
133.1	28.3
136.6	29.3
142.0	30.6
150.1	32.3
150.3	33.1

- (a) Taking the amount of wood pulp as the independent (or input) variable, find the estimated regression line.
 - (b) Predict the amount of newsprint produced in a year in which 146.0 million metric tons of wood pulp is produced.
 - (c) Taking the amount of newsprint as the input variable, find the estimated regression line.
 - (d) Predict the amount of wood pulp produced in a year in which 32.0 million metric tons of newsprint is produced.
8. It is believed that the more alcohol there is in an individual's bloodstream, the slower is that person's reaction time. To test this, 10 volunteers were given different amounts of alcohol. Their blood alcohol levels were determined as percentages of their body weights. The volunteers were then tested to determine their reaction times to a given stimulus. The following data resulted.

$x = \text{amount of alcohol in blood (percent)}$	$y = \text{reaction time (seconds)}$
0.08	0.32
0.10	0.38
0.12	0.44
0.14	0.42
0.15	0.47
0.16	0.51
0.18	0.63

- (a) Plot a scatter diagram.
 - (b) Approximate the estimated regression line by drawing a straight line through the data.
 - (c) What is the estimated regression line?
 - (d) Compare the lines in parts (b) and (c). Are their slopes nearly equal? How about the intercepts?
- Predict the reaction time for an individual (not one of the original volunteers) whose blood alcohol content is
- (e) 0.15
 - (f) 0.17
9. In [Example 12.2](#), suppose the eight training times had been chosen in advance. How do you think the decision as to the assignment of the eight workers to these training times should have been made?
10. In an experiment designed to study the relationship between the number of alcoholic drinks consumed and blood alcohol concentration, seven individuals having the same body size were randomly assigned a certain number of alcoholic drinks. After a wait of 1 hour, their blood alcohol levels were checked. The results were as follows.

Number of drinks	Blood alcohol level
0.5	0.01
1	0.02
2	0.05
3	0.09
4	0.10
5	0.14
6	0.20

- Draw a scatter diagram.
 - Find the estimated regression line, and draw it on the scatter diagram.
 - Predict the blood alcohol level of a person, of the same general size as the people in the experiment, who had 3 drinks 1 hour ago.
 - What if the person in part (b) had 7 drinks 1 hour ago?
11. The following data relate the per capita consumption of cigarettes in 1930 and men's death rates from lung cancer in 1950, for a variety of countries.

Country	1930 Per capita cigarette consumption	1950 Deaths per million men
Australia	480	180
Canada	500	150
Denmark	380	170
Finland	1100	350
Great Britain	1100	460
Iceland	230	60
The Netherlands	490	240
Norway	250	90
Sweden	300	110
Switzerland	510	250
United States	1300	200

- Determine the estimated regression line.
Predict the number of 1950 lung cancer deaths per million men in a country whose 1930 per capita cigarette consumption was
 - 600
 - 850
 - 1000
12. The following are the average scores on the mathematics part of the Scholastic Aptitude Test (SAT) for some of the years from 1994 to 2009.

Year	SAT Score
1994	504
1996	508
1998	512
2000	514
2002	516
2004	518
2005	520
2007	515
2009	515

Assuming a simple linear regression model, predict the average scores in 1997, 2006 and 2008.

13. Use the data of Prob. 3 in Sec. 3.7 to predict the IQ of the daughter of a woman having an IQ of 130.
14. Use the data of Prob. 6 in Sec. 3.7 to predict the number of adults on parole in a state having 14,500 adults in prison.
15. The following data relate the proportions of coal miners who exhibit symptoms of pneumoconiosis to the number of years of working in coal mines. Use it to estimate the probability that a coal miner who has worked for 42 years will have pneumoconiosis.

Years working	Proportion having pneumoconiosis
5	0
10	0.0090
15	0.0185
20	0.0672
25	0.1542
30	0.1720
35	0.1840
40	0.2105
45	0.3570
50	0.4545

12.4 ERROR RANDOM VARIABLE

We have defined the linear regression model by the relationship

$$Y = \alpha + \beta x + e$$

where α and β are unknown parameters that will have to be estimated and e is an error random variable having mean 0. To be able to make statistical inferences about the regression parameters α and β , it is necessary to make some additional assumptions concerning the error random variable e . The usual assumption, which we will be making, is that e is a normal random variable with mean 0 and variance σ^2 . Thus we are assuming that the variance of the error random variable remains the same no matter what input value x is used.